

# July 2026 work plan — the vacation month

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Source: owner call 2026-07-09 (transcript in `~/Documents/1-Projects/Jensen/ Misc - Transcripts/` ), processed same day. Decisions below marked **[dev]** were made by Nazar 2026-07-09 when this plan was drawn up.

*Dennis: this doc is for both of us. The prose says what's being built and why; the grey `code blocks` and file paths are developer blueprints — skip them freely, nothing in them changes the story around them.*

## The calendar constraint that shapes everything

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- Dennis is on vacation now, **back August 3**.
- Nazar works July ("taking it easy"), **on vacation from August 4**.
- Overlap  $\approx$  one day. So July's output must be **self-serve** — Dennis returns to an app he can onboard himself into (Danish, roles, polished maintenance/workshop flows, docs), and everything that needs the two of them together waits for mid-August.

## Asks sent to Dennis (can be answered from vacation, email-sized)

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1. ~~Forward the painter's price list~~ **RESOLVED 2026-07-09** — already in the shared folder (`Misc files from D/SIK_Jensen Priser 2026.xlsx` ), analyzed below. Three email-sized confirmations remain:
  - Prices are **ex moms**, right? (Assumed yes — B2B convention.)
  - Tier basis on mixed batches: 4 frames + 12 forks → frames at the 1–9 price and forks at the 10–19 price (**per part type**, assumed), or everything at the tier of the total piece count?
  - The list is from **SIK** — is SIK the painter going forward (the system's current default is Metacoat A/S)? Same painter renamed, a second painter, or a switch?
2. Nothing else blocks July. (Old-system export, role matrix details, invoicing walkthrough are all August items.)

## What the Jul 9 call actually changed

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Most of the call was demo/confirmation of shipped work. New substance:

| # | Item                                                                                                                                                                                                           | Status                                                                                          |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 1 | <b>Paint pricing remodel</b> — painter scrapped packages; every part (frame, fork, mudguards, chain guard, basket, sign, carrier...) priced individually with qty tiers; painter's numbers become item numbers | New decision, supersedes the Jul-2 std/svaj model<br><b>once shipped</b>                        |
| 2 | <b>i18n (translations) unparked</b> — Danish is a hard requirement (a workshop employee can't work in English); German is strategic for future sales                                                           | Was parked 2026-06-21, now active                                                               |
| 3 | <b>Roles requested</b> — "a login that says I can see everything, they can work on this and this and not into this"                                                                                            | Interim no-login answer in July; real login is a later milestone (M1)                           |
| 4 | <b>Old-data migration</b> — the daily "whose frame/battery is this?" lookup lives in the old system                                                                                                            | <b>[dev] Skipped for July</b> — needs export + Dennis's knowledge; whole track lands mid-August |
| 5 | <b>Global identifier search</b> — battery/frame number → bike → invoices, old-system parity                                                                                                                    | July, small                                                                                     |
| 6 | Housekeeping card rows should open the actual offending items                                                                                                                                                  | July, small                                                                                     |
| 7 | Mobile photo upload must work from a phone browser (Dennis fixes wrong AI-fetched images himself, from the field)                                                                                              | July, verify + fix                                                                              |
| 8 | Maintenance + workshop floor are what Dennis learns <b>first</b> in August                                                                                                                                     | July polish pass, in Danish                                                                     |
| 9 | Website bike-configurator, AI lead-gen agent                                                                                                                                                                   | Parked by Dennis himself ("lay the bottom first")                                               |

## July tracks, sequenced

Rough capacity: ~15 dev-days at vacation-adjacent pace.

### Week 1 — quick wins + the money item

- **Housekeeping drill-down** (~0.5 d): each row on the dashboard housekeeping fold links to a *filtered* list showing exactly the offending records (parts-without-origin filter etc.), per the

URL-driven-filters convention. Dennis asked for precisely this.

- **Mobile photo upload** (~0.5 d): verify on a real phone (camera → upload on part/bike), fix what's broken. Unblocks Dennis's August photo-cleanup habit.
- **External-services remodel — service\_orders + generic price lists** (~3.5–4 d, list in hand — analysis + full design below). [**owner decision 2026-07-09: "go all the way to service\_orders"**] — painting is the first *service type*, not a special case. Washing, priming, galvanizing, wheel building etc. reuse the same machine when they become real.
  - Template link: a template knows which of its parts are paintable → "cost to paint this bike" feeds cost-to-produce + margin (the 310→710 kr lesson: bad paint estimates ate the hotel-project margin).
  - The shipped std/svaj model ( `resolveLakSkus` ) stays authoritative until this ships — don't rip it out first, replace it in one cut.

### The painter's list, analyzed (SIK\_Jensen Priser 2026.xlsx, read 2026-07-09)

Source: `~/Documents/1-Projects/Jensen/Misc files from D/`, one sheet ( `Ark1` ). Column E ("PRIS pr. 1. juni 2026") is the **authoritative new per-piece price**; C/D are the painter's 2025 system prices (set / per piece) and H is older still — reference only. Item numbers carry stray trailing spaces/nbsp in the sheet — normalize on import.

**The catalog: 8 part types × 3 tiers (v/1–9, v/10–19, v/20 = 20+), 24 rows, DKK per piece:**

| Part (da)        | English           | Item nos ( <code>J. Jensen ...</code> ) | 1–9 | 10–19 | 20+ |
|------------------|-------------------|-----------------------------------------|-----|-------|-----|
| Stel             | Frame             | Stel1 / Stel10 / Stel20                 | 365 | 250   | 175 |
| Forgaffel        | Fork              | FG1 / FG10 / FG20                       | 70  | 60    | 50  |
| Lad              | Cargo bed         | Lad1 / Lad10 / Lad20                    | 185 | 130   | 110 |
| Skærm og stivere | Mudguards + stays | S1 / S10 / S20                          | 130 | 125   | 120 |
| Kædeskærm        | Chain guard       | KS1 / KS10 / KS20                       | 90  | 80    | 75  |
| Kurv             | Basket            | KU1 / KU10 / KU20                       | 185 | 160   | 140 |
| Skilt            | Sign              | Skilt1 / Skilt10 / Skilt20              | 90  | 80    | 75  |
| Bagagebærer      | Rear carrier      | Bag1 / Bag10 / Bag20                    | 185 | 165   | 150 |

**Continuity check that validates the reading:** the old frame price included the fork. New Stel + FG reproduces the old package at every tier —  $365+70 = 435$  ("Gammel" 1–9) ·  $250+60 = 310$  ·  $175+50 = 225$ . The painter split the package without moving the frame+fork total. Real increases

hide elsewhere: Kurv 20+ 95→140, Bag 10–19 130→165, Lad 20+ 95→110 — per-part itemization is exactly where the old model under-quoted (the 310→710 margin hit).

### The service model (final design, owner-approved 2026-07-09)

Painting generalizes to **outsourced per-part services**: a supplier takes a batch of part-units, prices from a supplier-issued tiered price list, the list gets revised every year or two, and the parts are physically away until received back. Washing, sandblasting/priming, galvanizing, powder-coat-vs-wet-paint, wheel building, external motor service all share the shape. Dennis also compares DK vs DE painters — so price lists hang off **suppliers**, enabling side-by-side estimates later. The parked website configurator prices paintable parts from this same layer.

### Pricing + vocab layer (all new):

```
service_types          -- WHAT kinds of work exist (controlled vocab)
  slug, name_da/en, is_active, sort_order,
  blocks_build bool    -- paint TRUE: parts physically away gate the
                        -- build floor; wash would be TRUE too
service_part_types     -- WHAT units work is done on (shared across types)
  slug, name_da/en, sort_order, is_active -- stel, forgaffel... (8 today)
service_price_lists    -- WHOSE prices, from WHEN — one row per REVISION
  supplier_id → suppliers, service_type_id,
  name ("SIK priser 2026"), currency, effective_from, is_current
service_price_items    -- the numbers — one row per list line (24 today)
  price_list_id, service_part_type_id,
  supplier_item_no ("J.Jensen Stel10"), tier_min, tier_max (NULL=open),
  price
```

**Multi-country / multi-currency (owner requirement 2026-07-09):** service suppliers live anywhere (DK painter today, DE painter tomorrow, `suppliers.country_code` already exists) and each price-list revision carries its own `currency`. Handling mirrors purchasing exactly: estimates convert non-DKK lists to DKK via the shared `lookupFxRate` (ECB, same as stock-adjust foreign cost); at send, the item snapshot freezes `unit_price + currency + fx_rate_to_dkk` — the `purchase_order_lines` pattern — so a sent order's DKK cost basis never drifts with the exchange rate. Cross-supplier comparison ("paint this batch in DK vs DE") converts both sides to DKK at today's rate; the transport-cost leg of that comparison (Dennis: "logistics balance it out") is a future estimator refinement, noted not built.

**Order layer — `paint_orders` PROMOTED to `service_orders`** (owner: "go all the way"):

```

service_orders      -- rename of paint_orders
  order_no          -- next_document_number per type: paint keeps
                    -- 'paint_order' → PNT-YYYY-NNNN; wash gets its own
  service_type_id, supplier_id, status (planned → sent → at_supplier →
  received_back / cancelled), sales_order_id?, notes, dates
service_order_items -- NEW: the commercial lines
  service_order_id, service_part_type_id, qty,
  color_id (nullable FK colors – paint uses it, wash ignores; a real
  column, NOT jsonb – FK integrity + pickers),
  snapshot at send: supplier_item_no, unit_price, currency,
  fx_rate_to_dkk (frozen, purchase_order_lines pattern)
service_order_bikes -- rename of paint_order_bikes: bike linkage for
                    -- the at-supplier gate + traceability. Existing
                    -- per-line colour/scope columns stay as LEGACY
                    -- (read-only) so sent/received history keeps its
                    -- frozen cost basis; new orders use items.

```

### Rules carried over from the app's conventions:

- Tier resolution is live while `planned` (estimates track the current list), **frozen at send** — snapshot `supplier_item_no` + unit price onto the item, same rule as `fx_rate_to_dkk`. A new price list never rewrites a sent order.
- Tier basis: per part type's total qty on the order (4 frames + 12 forks → frames tier 1–9, forks tier 10–19) — pending Dennis's email confirmation.
- At-supplier gate generalizes: bike blocked iff on an open service order whose `service_types.blocks_build` — `at-painter.ts` becomes `src/lib/services/at-supplier.ts`, same call sites.
- One pricing brain: `src/lib/services/pricing.ts` → `resolvePrice(supplierId, serviceTypeId, partTypeId, qty, date)` — used by the order estimator, template cost-to-paint, and later the configurator/quotes.

**Price changes (the "very updatable" requirement):** `/admin/services` — per supplier × service type, the current list renders as a part-type × tier grid with revision history. A change is never edit-in-place: **new revision** via duplicate-and-edit or **upload the supplier's next xlsx** → parsed → **diff preview** ("Stel 10–19: 250 → 265") → confirm with effective date; old revision archives ( `is_current` flip, the bike\_templates versioning pattern). Yearly price bump = 5-minute clerical task.

**Nav & routes — per-service-type surfaces (owner decision 2026-07-09):** there is deliberately NO unified "Service orders" page planned. Each service type gets its **own nav item and list route** when it becomes real — "Paint orders" at `/paint-orders` today; "Sandblasting" / "Washing" would each get their own entry later. All of them render the same shared components (list, detail, items editor, status flow) parameterized by `service_type`; only the

surface is per-type, the machine underneath is one. This fits how the workshop thinks ("send frames to the painter", "send parts to washing" — different errands), keeps the two-nav sync rule manageable (add one item per real service), and means the paint routes never need a rename at all.

### Cutover mechanics:

- Routes + nav: `/paint-orders` / "Paint orders" stay, permanently — per-type surfaces above. Tables and code go generic underneath.
- Scope → items mapping for open `planned` orders: `std` = Stel + FG; `svaj` extras = S + Skilt + Bag (or Lad where fitted). Sent/received orders keep legacy lines untouched.
- JP-lak service SKUs retire; 7 files reference them (coverage, readiness, pick-list, reorder draft-PO, scope.ts + 2 paint UI files) — each either drops its exclusion hack or switches to the pricing lib.
- 14 files reference the old table names; 14 route files under `/paint-orders` — bounded rename, one cut, browser-verified per the discipline.
- Seed migration: `service_types` (lakering, blocks\_build), 8 part types, the SIK 2026 list (24 items, currency DKK, effective 2026-06-01). ⚠️ Confirm supplier identity: the file says **SIK**, CLAUDE.md's default painter is **Metacoat A/S** — create/link the right supplier row and set the paint default accordingly (ask Dennis alongside the VAT/tier questions).
- VAT: prices assumed **ex moms** (confirm with Dennis, see asks).

## Weeks 2–3 — the two big tracks

- **i18n, whole app to Danish** (~4–5 d) [**dev decision: whole-app scope**]: next-intl foundation; worker screens first (keyed off `worker_language` — covers the employee who can't work in English), then sweep every module keyed off `app_language`; **de locale scaffolded but untranslated** (German has no user yet). Customer-facing documents keep their own per-document `language` — unchanged.
- **Phone→ticket pipeline v1** (~4 d) — deep dive below. Built harness-first; Twilio wired last.

## Week 4 — Dennis-return enablers

- **Interim device roles** (~1 d) — design below [**dev decision: cookie, no PIN**].
- **Global identifier search** (~1 d): one search box (header or `/search?q=`) hitting `bike_identifiers` (exact/prefix), bikes (frame number), parts (SKU/name trigram), organizations, invoices (number) — grouped results, each row linking home. Old-system parity for the daily lookup.
- **Maintenance + workshop floor polish pass** (~1 d): walk both flows in Danish on a phone, fix rough edges; QR codes printable; this is Dennis's first-touch surface in August.

- **Handover notes** (~0.5 d): a short "what changed in July + how to test it" doc for Dennis's return, plus updating this plan's checkboxes.

## Deep dive: phone-call → ticket pipeline

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**What this is, in one sentence:** a customer calls the workshop, nobody can pick up, they leave a voicemail — and the system turns that voicemail into a draft maintenance ticket (transcribed, customer and bike identified where possible) waiting for review. Nothing auto-sends to customers in v1; it just saves the "who called about what" typing.

The parked design (CLAUDE.md → Parked ideas) survives review with three adjustments: build order, transcription provider, and an explicit "harness-first" v1 — we build and tune it by uploading hand-recorded voicemails first, and connect a real phone number only at the very end.

### Build order — harness first, telco last

~80 % of the pipeline needs no phone number at all. Everything from "audio file exists" onward is provider-independent:

1. **Migration:** `calls` table (recording path, caller number, duration, language, transcript jsonb w/ timestamps, summary, extraction payload, candidate bikes, pipeline status enum `received → transcribed → extracted → matched → ticketed / failed`, nullable `ticket_id`, raw provider payload jsonb for replay). Plus a `fleet_number` row in `bike_identifier_types` (customers' own numbering — "bike 25" — big match-rate win for municipalities).
2. **Processing pipeline** ( `src/lib/calls/` ): `transcribe → extract → match → draft ticket`. Each stage writes its output + advances status, so a failure is resumable and inspectable.
3. **Test harness UI** ( `/admin/calls` ): list of calls + **"Upload a voicemail"** (any audio file → Supabase Storage → `calls` row) + per-call detail: audio player, transcript, extraction JSON, matching candidates, "Process now" button, "Create ticket / attach to existing" review actions. Record fake voicemails on a phone in Danish and English; iterate on extraction quality with zero telephony cost.
4. **Shadow-mode ticketing:** pipeline creates tickets with `source='phone'` + a "from phone — review me" banner; deterministic matching attaches a bike only when exactly one candidate survives, else stores candidates for the tech to confirm. Measure match accuracy on the call rows before trusting anything.
5. **Twilio wiring (last, ~1 d):** buy DK number, conditional forwarding from the workshop number (busy/no-answer), TwiML voicemail flow with the bilingual da/en "call is recorded" announcement (GDPR), webhook route → store call row → **fetch recording into Supabase Storage (EU) → delete from Twilio immediately**. Processing trigger: inline if it



fits the function budget, else Vercel cron each 5 min + the manual "Process now" button (which exists anyway from the harness).

## Provider verdicts [dev decisions 2026-07-09]

| Layer         | Choice                                                           | Why / mitigations                                                                                                                                                                                                                                                                                                                                                      |
|---------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Telephony     | <b>Twilio</b>                                                    | Best docs + dev speed for a solo build. GDPR mitigations: recordings pulled to Supabase EU and deleted from Twilio at webhook time; recording announcement covers consent. EU-native fallback if this ever chafes: 46elks (Swedish, DK numbers, voice+SMS, EU residency) — swap cost is small because the provider surface is one webhook route + one recording fetch. |
| Transcription | <b>Azure Speech-to-Text, EU region</b> (or Azure-hosted Whisper) | Plain OpenAI Whisper API has no EU residency guarantee. Azure: Danish quality good, EU region explicit, per-channel transcription ready for the dual-channel v2. New env secrets: <code>AZURE_SPEECH_KEY</code> / <code>AZURE_SPEECH_REGION</code> .                                                                                                                   |
| Extraction    | <b>Claude API</b> , <code>claude-haiku-4-5</code>                | Structured output via tool use: caller, org, callback number, bike clues, problem, urgency, language. Escalate to Sonnet only if Danish extraction quality disappoints. Env: <code>ANTHROPIC_API_KEY</code> . Matching stays deterministic code, never the model.                                                                                                      |
| SMS ack       | <b>GatewayAPI</b> (Danish)                                       | Unchanged from the parked design; v1 shadow mode sends nothing — SMS starts when tickets auto-create for real.                                                                                                                                                                                                                                                         |
| Storage       | <b>Supabase Storage (EU)</b>                                     | Already the stack; audio retention ~90 days (cron delete), transcript/summary live on the ticket.                                                                                                                                                                                                                                                                      |

Secrets live in env, operational config (the workshop number, forwarding target) in `app_settings` — `workshop_phone` already exists (migration 55) per the config-vs-secrets rule.

## Matching (unchanged from parked design — deterministic, in order)

1. Caller ID → `contacts.phone` → organization.
2. Spoken org name → trigram on `organizations.legal_name` .
3. Spoken frame / QR / **fleet number** → `bike_identifiers` exact.



4. Else: owner org's fleet filtered by colour/type clues. Attach bike iff exactly one candidate survives; otherwise store candidates on the call for the ticket banner.

## GDPR non-negotiables (v1)

Recording announcement (da/en) before the beep · audio retention ~90 days, then deleted (transcript + summary persist on the ticket) · DPAs: Twilio, Microsoft, Anthropic · all storage EU · caller SMS/ack deferred until out of shadow mode.

## Cost (unchanged estimate)

< 2 kr per 5-minute call end-to-end + ~50 kr/month for the number.

## Interim roles — device-role cookie [dev decision: no PIN]

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**What you'll experience:** the workshop tablet shows only workshop screens (floor, maintenance, bikes, parts); your own devices see everything. No passwords, no accounts yet — each *device* is marked once as "owner" or "workshop", and that's it until real login arrives.

Dennis's ask, minus the login nobody wants to build twice:

- **device\_role signed httpOnly cookie**, values `owner | workshop`. Absent cookie = `owner` (desktops unaffected; the workshop tablet gets flipped once).
- **Owner-only switcher** at `/admin/device-role` (System tile): "This device is: Owner / Workshop".
- **One helper:** `can(role, capability)` in `src/lib/access/` — nav filtering (BOTH `app-sidebar.tsx` and `mobile-nav.tsx` — the two-nav sync rule) and layout-level route guards read only this.
- **v1 matrix (coarse):** `workshop` sees Daily ops (Workshop floor, Maintenance, Bikes, Parts, Templates) and lands on `/work`; the Orders & commercial group, Admin, Dashboard money band are hidden and their routes redirect. Cost fields inside parts/templates are a v1.1 refinement — Dennis defines the fine matrix in August ("they can work on this and this and not into this" — his words, his call).
- **Explicitly not security** — Vercel SSO stays the outer wall; this is ergonomics so a shared tablet doesn't wander into invoices. Flag: the workshop tablet needs a one-time Vercel SSO login (or a bypass link) — solve at rollout, not in code.
- **Forward-compatible by construction:** the role enum + `can()` are exactly M1's shape; auth later swaps the role source (cookie → session/ `profiles.role`) and nothing else moves. The agreed M1 trigger (first real invoice) still stands.

## August-with-Dennis backlog (documented now, not July work)

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- **Old-data migration** — export + schema mapping + import + verify, with Dennis's knowledge of the old system in the room. (Skipped in July by decision; see table above.)
- **Invoicing parity workshop** — his walk-in flow: draft → customer arrives → issue → mark paid → push lands in e-conomic **with a "paid" remark** so bookkeeping just books it. The paid-remark on the voucher text is a small addition to `push-invoice.ts` once the flow is agreed.
- **e-conomic production cutover** — grant token for the real agreement (~end of July per 3E notes), re-run Test connection, confirm journal / account / VAT / payment-terms numbers with the revisor, first real push.
- **Supplier-email go-live** — Dennis fills the 18 missing supplier emails (housekeeping card), `orders@valent.dk` alias created, untick test mode.
- **Role matrix refinement** — per-capability decisions on the cookie roles.
- **Service-agreement-on-bike verification** — needs real bikes in the system (post-migration / post data-entry).
- **Action-items dashboard fine-tuning together** — his "what needs to be fixed/checked" wishlist, refined against real data.

## Parked (by Dennis's own call: "lay the bottom first")

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- Website bike-configurator that talks to the app (customer designs a bike on the homepage, colour → closest RAL, paintable parts + live price, "send me an offer").
- AI lead-gen agent (find nearby SMBs moving offices / hotels / municipalities; auto-draft outreach with the configurator link).
- Both belong to the sales track — revisit after the system is the daily workhorse and (Dennis's framing) once debt is down; earliest next year.